

Name: SRINJOY CHAKRABORTY

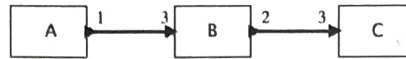
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CS637 Embedded and Cyber-Physical Systems

Homework Assignment 1

Deadline: September 11, 2026

Problem 6. (10 points)

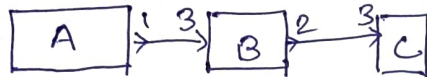
Consider the following SDF model:



The numbers adjacent to the ports indicate the number of tokens produced or consumed by the actor when it fires. Answer the following questions about this model.

(a) Let q_A , q_B , and q_C denote the number of firings of actors A , B , and C , respectively. Write down the balance equations and find the least positive integer solution.

(b) Find a schedule for an unbounded execution that minimizes the buffer sizes on the two communication channels. What is the resulting size of the buffers?

Figure

(a)

No. of firings are $\rightarrow q_A, q_B, q_C$

Edge A to B $\Rightarrow 1 \cdot q_A = 3 \cdot q_B$ — (i)

Edge B to C $\Rightarrow 2 \cdot q_B = 3 \cdot q_C$ — (ii)

Solving (i) & (ii)

$$q_A = 3q_B, \quad q_C = \frac{2}{3}q_B$$

The least possible solutions are:-

$$q_B = 3$$

$$\therefore q_A = 9, \quad q_C = 2$$

(b)

We need to minimize the buffer size.

$\therefore$ Fire the actor, the exact moment it has enough tokens.

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The overall token count change:-

① Fire A $\rightarrow$ 3 times. AB buffer = 3
BC buffer = 0

② B now has enough, fire B $\rightarrow$ 1 time.
AB = 0
BC = 2

③ As C needs 3 tokens, but has 2, fire A $\rightarrow$ 3 times.
B $\rightarrow$ 1 time.
 $\therefore$ AB = 0
BC = 4

④ Now C has enough. Here C is fired 1 time.
AB = 0
BC = 1

⑤ Again fire A $\rightarrow$ 3 times, B $\rightarrow$ 1 time.
AB = 0
BC = 3

⑥ Fire C 1 time. AB = 0
BC = 0

Schedule $\rightarrow$ AAABAAABCAAAABC.Buffer SizeBuffer A $\rightarrow$ B : 3Buffer B $\rightarrow$ C : 4